Auteur: Xander De Blauwer
Studierichting: Informatica
Oefeningenreeks 5I nr. 1.6

$$f(x) = \frac{2}{3}x^{\frac{3}{2}} - \frac{1}{2}x^{\frac{1}{2}} \text{ op } [1, 4]$$

$$f'(x) = x^{\frac{1}{2}} - \frac{1}{4}x^{-\frac{1}{2}} = \frac{4x-1}{4\sqrt{x}}$$

$$S = \int_1^4 \sqrt{\frac{16x^2+8x+1}{16x}}dx = \int_1^4 \frac{4x+1}{4\sqrt{x}}dx$$

$$= \left[\frac{2}{3}x^3 2\right]_1^4 + \frac{1}{2}\left[\sqrt{x}\right]_1^4$$

$$= \frac{2}{3}\cdot 8 - \frac{2}{3} + 1 - \frac{1}{2} = \frac{31}{6}$$

5I-1.6-xander.de.blauwer.INF.png

References